AIML Lecture 4 MCQ

Machine Learning Pipeline & Data Collection

1. What is the primary purpose of a Machine Learning Pipeline?
   a) To manually analyze raw data for insights.
   b) To solely deploy machine learning models.
   c) **To transform raw data into valuable predictions using a sequence of steps.**
   d) To collect data from various sources.
2. Which of the following is NOT a key stage in the Machine Learning Pipeline mentioned in the lecture?
   a) Problem Definition
   b) Data Collection
   c) **Marketing Strategy**
   d) Data Preprocessing
3. Which type of data is typically stored in a tabular format and is easily searchable?
   a) Unstructured data
   b) Textual data
   c) **Structured data**
   d) Audio data
4. An email or a social media post would be considered an example of what type of data?
   a) Structured data
   b) Tabular data
   c) Numerical data
   d) **Unstructured data**
5. Which of the following is explicitly mentioned as a source of data for machine learning projects?
   a) Satellite imagery
   b) **APIs**

c) User interviews
d) Legacy hard drives

6. Which tool is specifically listed for web scraping in the lecture?
   a) Postman
   b) **BeautifulSoup**
   c) Apache NiFi
   d) Python Requests

7. ETL tools like Apache NiFi and Talend are primarily used for what purpose in the data acquisition phase?
   a) Model training
   b) Data visualization
   c) **Extracting, Transforming, and Loading data**
   d) Deploying models

8. What is the first step in a Machine Learning Project, as outlined in the lecture?
   a) Data Collection
   b) Data Preprocessing
   c) Model Training
   d) **Problem Definition**

9. Why is problem definition a crucial first step in a Machine Learning project?
   a) It determines the budget for the project.
   b) It specifies the exact algorithms to be used.
   c) **It guides the entire process from data collection to model deployment by clarifying the objective.**
   d) It identifies the hardware requirements.

10. A dataset containing customer names, addresses, and purchase history in a SQL database would typically be classified as:
    a) Unstructured data
    b) Semi-structured data
    c) **Structured data**
    d) Time-series data

Data Preprocessing

1. What is the main reason raw data needs to be preprocessed before using it in a machine learning model?
   a) Raw data is always too small for models.
   b) **Raw data is often messy and needs to be cleaned, fixed, and prepared.**
   c) Raw data is not compatible with modern operating systems.
   d) Raw data must first be converted into an image format.

2. If a temperature value is missing in a dataset, which of the following is a potential solution mentioned for handling missing data?
   a) Convert it to a string "missing".
   b) **Fill with the average (mean).**
   c) Automatically predict it with another ML model.
   d) Ignore the missing value during training.

3. When handling missing values, if "too many values" are missing in a row, what is one suggested solution?
   a) Fill with the mode.
   b) Impute using a complex algorithm.
   c) **Drop the row.**
   d) Fill with the previous valid value.

4. Why do machines need categorical data like "Yes"/"No" to be converted into numbers?
   a) To make the dataset smaller.
   b) To improve human readability.
   c) **Machines work with numbers, not words.**
   d) To encrypt the data for security.

5. Which encoding method converts a categorical feature like "Country" (e.g., UK, India) into separate columns, where each column represents a country with binary values (0 or 1)?
   a) Label Encoding
   b) **One-Hot Encoding**
   c) Target Encoding
   d) Ordinal Encoding

6. Which encoding technique replaces categories with the mean of the target variable for each category?
   a) Label Encoding

b) One-Hot Encoding
c) Binary Encoding
d) **Target Encoding**

7. What is the primary purpose of data scaling in machine learning?
   a) To reduce the number of features.
   b) To increase the interpretability of features.
   c) **To ensure all features are treated equally, preventing larger numbers from dominating.**
   d) To convert numerical data to categorical data.
8. If "Age" ranges from 10-80 and "Marks" range from 0-10, which scaling method would rescale values between 0 and 1, ensuring they have a similar range?
   a) Standardization (Z-score)
   b) Robust Scaling
   c) **Min-Max Scaling**
   d) Log Transformation
9. Which scaling method makes data follow a standard scale with a mean of 0 and a standard deviation of 1?
   a) Min-Max Scaling
   b) **Standardization (Z-score)**
   c) Normalization
   d) Decimal Scaling
10. A dataset containing "Salary=10" for an individual who is expected to have a much higher salary is an example of what type of dirty data?
    a) Incomplete data
    b) Inconsistent data
    c) **Noisy data**
    d) Intentional missing data
11. If a database entry shows "Age=42" but "Birthday=03/07/2010", this represents which type of data problem?
    a) Noisy data
    b) Incomplete data
    c) **Inconsistent data**
    d) Disguised missing data

12. The practice of filling in "Jan. 1" as everyone's birthday to avoid recording missing data is an example of what?
    a) Noisy data
    b) Inconsistent data
    c) Random imputation
    d) **Intentional (e.g., disguised missing data)**
13. Which of the following is NOT a step mentioned under "Steps in Data Preprocessing"?
    a) Handling Missing Data
    b) Converting Categorical Data
    c) **Feature Selection**
    d) Scaling the Data
14. In the context of the BrightFuture Academy problem (Tute_4), if the 'LessonsCompleted' column has missing values, and it's a numerical feature, what would be a suitable common method to handle them?
    a) Drop all rows with missing 'LessonsCompleted'.
    b) Fill with a string like "N/A".
    c) **Fill with the average (mean) of 'LessonsCompleted'.**
    d) Convert the column to a categorical feature.
15. For the 'Result' column in the BrightFuture Academy dataset (Pass/Fail), which encoding technique would be most appropriate if it's the target variable?
    a) One-Hot Encoding
    b) Target Encoding
    c) **Label Encoding (e.g., Pass → 1, Fail → 0)**
    d) Binary Encoding

Feature Engineering

1. What is the primary goal of feature engineering?
    a) To reduce the number of samples in the dataset.
    b) To speed up the data collection process.
    c) **To transform raw data into meaningful features that improve model performance.**
    d) To evaluate the performance of machine learning models.

2. According to the lecture, good features can make simple models powerful, while poor features can render complex models useless. When does feature engineering typically occur in the ML pipeline?
   a) Before data collection.
   b) After model training.
   c) **After data preprocessing and before model training.**
   d) During problem definition.

3. In the student data example (Hours Studied, Attendance, Sleep Hours, Result), what is "Result" considered?
   a) A feature
   b) A hyperparameter
   c) **The target variable**
   d) An outlier

4. Why is feature engineering necessary if machines are supposed to "learn" from raw data?
   a) Machines can only understand visual data.
   b) **Raw data often contains unhelpful, confusing, or hidden information that needs to be made useful.**
   c) Feature engineering simplifies the deployment process.
   d) It makes the data less clean for the machine.

5. Which of the following is a step involved in Feature Engineering, as described in the lecture?
   a) Building the final machine learning model.
   b) **Choosing the best features.**
   c) Deploying the model to production.
   d) Monitoring model performance.

6. What is the purpose of the Variance Threshold feature selection technique?
   a) To select features with high correlation to the target.
   b) To combine features into principal components.
   c) **To remove features that do not vary much across samples and thus carry little information.**
   d) To identify non-linear relationships.

7. A feature with zero or near-zero variance is likely constant or almost constant. Why is such a feature not helpful for prediction?

a) It consumes too much memory.
b) It increases computational complexity.
c) **It provides little to no discriminative information to the model.**
d) It leads to overfitting.

8. When is Variance Threshold particularly useful?
   a) For categorical data.
   b) When the target variable has a non-linear relationship with features.
   c) **As a preprocessing step for high-dimensional numerical data.**
   d) For selecting features with strong linear correlations.

9. What does Mutual Information measure between a feature and the target variable?
   a) The linear relationship.
   b) The inverse relationship.
   c) **The amount of information shared, capturing any kind of dependency (linear or non-linear).**
   d) The difference in variance.

10. Unlike correlation, Mutual Information can capture what kind of dependency?
    a) Only linear dependencies.
    b) Only inverse dependencies.
    c) **Any kind of dependency (linear or non-linear).**
    d) Only monotonic relationships.

11. SelectKBest (or SelectPercentile) allows selection of the top K features based on what?
    a) Their individual variance.
    b) Their computation time.
    c) **A scoring function, which can be a statistical test like chi-square or mutual information.**
    d) Their average value.

12. Which feature selection method uses a machine learning model itself to determine the importance of features during training?
    a) Variance Threshold
    b) Mutual Information
    c) SelectKBest
    d) **Model-Based Selection (e.g., Random Forest)**

13. What attribute in models like Random Forest allows access to feature importance scores for Model-Based Selection?
    a) `.coefficients`
    b) `.weights`
    c) **`.feature_importances_`** (Implied from the "Access .feature importances attribute" bullet, which is the common way this is done in ML libraries.)
    d) `.feature_scores`

14. Which category of feature selection methods selects features based on statistical tests and intrinsic properties of the data, independent of any machine learning algorithm?
    a) Wrapper Methods
    b) Embedded Methods
    c) **Filter Methods**
    d) Dimensionality Reduction

15. What is a key characteristic of Filter Methods for feature selection?
    a) They are computationally expensive.
    b) They are prone to overfitting.
    c) They are model-specific.
    d) **They are fast and computationally efficient, useful for preprocessing large datasets.**

16. Which feature selection approach evaluates different combinations of features using a predictive model, where performance (e.g., accuracy) determines which features to keep?
    a) Filter Methods
    b) **Wrapper Methods**
    c) Embedded Methods
    d) Dimensionality Reduction

17. What is a potential drawback of Wrapper Methods compared to Filter Methods?
    a) Less accurate.
    b) **Computationally expensive.**
    c) Not tailored to a specific model.
    d) Less prone to overfitting.

18. In which feature selection method is the selection process integrated within the model training, where the model itself selects important features during learning?
    a) Filter Methods
    b) Wrapper Methods
    c) **Embedded Methods**
    d) Univariate Selection

19. Which type of feature selection method offers a balance between performance and computational cost?
    a) Filter Methods
    b) Wrapper Methods
    c) **Embedded Methods**
    d) Hybrid Methods

20. What is the primary idea behind Dimensionality Reduction?
    a) To increase the number of features.
    b) To select the most relevant features directly.
    c) **To combine or remove features, keeping only what's essential by transforming them.**
    d) To evaluate model performance.

21. What is a key difference between Feature Selection and Dimensionality Reduction?
    a) Feature selection creates new features, while dimensionality reduction removes existing ones.
    b) Dimensionality reduction only works for numerical data, while feature selection works for all data types.
    c) **Feature selection picks a subset of original features, while dimensionality reduction transforms features into a new, smaller set.**
    d) Feature selection aims to reduce model complexity, while dimensionality reduction aims to improve model accuracy.

22. In the context of the BrightFuture Academy problem (Question 01, Tute_4), if 'TimeSpentPerWeek' and 'LessonsCompleted' are highly correlated, and both are predictive, which feature selection technique might combine them into a single, more abstract feature?
    a) Variance Threshold

b) Model-Based Selection
c) Mutual Information
d) **Dimensionality Reduction (e.g., PCA)**

23. For the dataset in Tutorial Question 01, if we want to create a new feature called `EngagementLevel = LessonsCompleted / (TimeSpentPerWeek + 0.0001)`, what process is this an example of?
    a) Data Normalization
    b) Feature Selection
    c) **Feature Engineering**
    d) Dimensionality Reduction

24. Considering the features in Tutorial Question 01's dataset (`Age`, `HoursStudied`, `Attendance`, `SleepHours`, `LessonsCompleted`, `TimeSpentPerWeek`, `AvgQuizScore`, `PreviousAttempts`, `EngagementLevel`), which feature selection method would be most appropriate if we suspect non-linear relationships with the `Result` target?
    a) Variance Threshold
    b) **Mutual Information**
    c) SelectKBest with chi-square
    d) Principal Component Analysis (PCA)

25. If the BrightFuture Academy dataset has many features and we want a computationally efficient way to reduce the number of features before training, without considering a specific model yet, which general category of feature selection would be most suitable?
    a) Wrapper Methods
    b) Embedded Methods
    c) **Filter Methods**
    d) Model-Based Selection

Model Introduction, Training & Evaluation

1. What is a Machine Learning Model described as?
    a) A database for storing raw data.
    b) A graphical user interface for data visualization.
    c) **A recipe or formula the computer learns to make predictions.**
    d) A tool for data preprocessing.

2. Which of the following is NOT a factor to consider when selecting a machine learning model?
   a) What is the problem type (Classification, Regression, Clustering)?
   b) How much data do you have?
   c) Do you need the model to be explainable?
   d) **The color scheme of the user interface.**

3. During model training, the data is typically split into which two main sets?
   a) Input set and Output set
   b) Development set and Production set
   c) **Training set and Test set**
   d) Validation set and Deployment set

4. What is the typical proportion of data allocated to the training set during data splitting?
   a) 20%
   b) 50%
   c) **80%**
   d) 100%

5. What is the key difference between a "parameter" and a "hyperparameter" in a machine learning model?
   a) Parameters are defined by the user, while hyperparameters are learned by the model.
   b) Hyperparameters are always numerical, while parameters can be categorical.
   c) **Parameters are learned by the model from the data, while hyperparameters are settings defined before training.**
   d) There is no significant difference; the terms are interchangeable.

6. Why is hyperparameter tuning important for model performance?
   a) To decrease the amount of training data needed.
   b) **To improve model performance, avoid overfitting/underfitting, and make the model more efficient.**
   c) To change the underlying machine learning algorithm.
   d) To reduce the number of features in the dataset.

7. Which of the following is NOT listed as an example of a hyperparameter?
   a) Learning rate

b) Number of layers in a neural network
c) Regularization strength
d) **Weights in linear regression** (Weights are parameters learned by the model)

8. What is the primary reason for using Cross-Validation?
   a) To speed up the model training process.
   b) To ensure the model is always overfitting.
   c) **To check if the model performs consistently and provides a more reliable performance estimate than a single train/test split.**
   d) To perform feature selection.

9. In K-Fold Cross-Validation, if K=5, how many times is the model trained and tested, with a different part of the data used for testing each time?
   a) 1 time
   b) 2 times
   c) **5 times**
   d) K-1 times

10. What is a key benefit of Cross-Validation?
    a) It guarantees perfect accuracy.
    b) It allows the model to memorize the training data.
    c) **It provides a more reliable performance estimate by reducing the impact of a specific data split.**
    d) It eliminates the need for data preprocessing.

11. What is the main purpose of Model Evaluation?
    a) To retrain the model with new data.
    b) To select the best hyperparameter values.
    c) **To assess if the model is accurate and generalizes well to new (unseen) data.**
    d) To deploy the model to production.

12. A model that "learns too much, including noise and outliers" from the training data and performs poorly on test data is experiencing what?
    a) Underfitting
    b) **Overfitting**
    c) Optimal fitting
    d) Generalization

13. Which of the following is a sign of overfitting?
    a) Model performs poorly on both training and test data.
    b) Model is too simple to capture patterns.
    c) **Model performs well on training data, but poorly on test data.**
    d) Model cannot make any predictions.

14. What happens when a model is "too simple" and "doesn't learn enough from the training data," leading to poor performance on both training and test data?
    a) Overfitting
    b) **Underfitting**
    c) Optimal fitting
    d) Bias-variance trade-off

15. Which of the following is NOT a suggested way to fix underfitting?
    a) Add more features.
    b) Use a more complex model.
    c) **Reduce model complexity.** (This is for overfitting)
    d) Increase the number of training iterations.

16. What is `K-Fold` an example of?
    a) A feature engineering technique.
    b) A data scaling method.
    c) A dimensionality reduction technique.
    d) **A Cross-Validation technique.**

17. If BrightFuture Academy's model for predicting student failure performs 95% accurately on its training data but only 60% on new student data, what is the most likely issue?
    a) Underfitting
    b) **Overfitting**
    c) Lack of data preprocessing
    d) Correct model generalization

18. For the problem in Tutorial Question 02, where the institute wants to group students into meaningful categories based on interaction data without labeled outcomes, what type of ML problem is this?
    a) Classification
    b) Regression
    c) **Clustering** (Although not explicitly stated in Tute_4, the phrase "group

students into meaningful categories or clusters" strongly implies clustering, and it's a fundamental ML problem type mentioned in lecture 4, Source).
d) Reinforcement Learning

Deployment & Post-Deployment

1. What is the process of making a trained and tested machine learning model available for real-world use called?
   a) Model training
   b) Feature engineering
   c) **Model deployment**
   d) Data collection

2. What is one of the primary goals once a model makes accurate predictions, regarding its deployment?
   a) To keep it exclusively in the developer's notebook.
   b) To only use it for internal testing.
   c) **To make it available for use by other people (e.g., via a website app).**
   d) To document the code and archive it.

3. Which of the following is a crucial activity in the post-deployment phase of a machine learning model?
   a) Discarding the training data.
   b) **Monitoring model performance (accuracy may drop over time).**
   c) Redefining the problem statement.
   d) Stopping all further updates to the model.

4. Why is "monitoring model performance" essential post-deployment?
   a) To ensure the model is not too fast.
   b) To collect new data for a different project.
   c) **Model accuracy may drop over time due to data drift or changing patterns.**
   d) To justify the initial investment in the model.

5. If a deployed model starts making less accurate predictions over time, what is one recommended action?
   a) Redeploy the exact same model.
   b) Ignore the performance drop.

c) **Update the model with new data (retraining).**
d) Increase the number of features.

Advanced Preprocessing & Feature Engineering Scenarios (Harder)

1. Consider the dataset in Tutorial Question 02. If the 'TimeSpentPerWeek' feature (which is numerical) shows a highly skewed distribution (e.g., many students spend very little time, but a few spend exceptionally long hours), which scaling method would be more robust to outliers and make the data follow a standard scale?
   a) Min-Max Scaling (Sensitive to outliers)
   b) **Standardization (Z-score, as it centers data around 0 and scales by std dev, making it robust if combined with outlier handling or if the outlier itself is transformed first, though Min-Max might compress outliers into the range making them appear less impactful, Standardization is generally preferred for normal distribution assumptions).**
   c) Label Encoding
   d) One-Hot Encoding

2. In Tutorial Question 02, if 'PreferredLearningStyle' is a categorical variable with values like "Visual", "Auditory", "Kinesthetic", and there's no inherent order, which encoding technique is most appropriate?
   a) Label Encoding (Imposes arbitrary order)
   b) Target Encoding (Requires a target variable, which is missing here)
   c) **One-Hot Encoding**
   d) Ordinal Encoding

3. For the dataset in Tutorial Question 02, if a new feature `Learning Efficiency = Time Spent Per Week / Avg Quiz Score` is created, and it contains division by zero (or a very small number leading to large values), what preprocessing step might be necessary for this *new* feature?
   a) One-Hot Encoding
   b) **Scaling (e.g., Standardization or Min-Max) to handle potential large ranges and extreme values.**
   c) Dropping the feature entirely.
   d) Converting it to categorical.

4. For a dataset similar to BrightFuture Academy's, if the `PreviousAttempts` feature (number of times a student tried an exam) ranges from 0 to 5, and `AvgQuizScore` ranges from 0 to 100, which scaling method would ensure both features contribute equally to a distance-based algorithm without changing their distribution shape?
   a) **Standardization (Z-score scaling centers the data and scales by standard deviation, which can be beneficial for algorithms sensitive to feature magnitudes, preserving the relative distances/distributions better than min-max when range differences are vast like this).**
   b) Min-Max Scaling (While it brings values to, it may still be sensitive to outliers if they exist in `AvgQuizScore` and simply squeezes them into the new range)
   c) Label Encoding
   d) Normalization to unit length

5. If a student's `Age` is missing (e.g., Occupation=" " from source) in a dataset, and age is critical for predicting learning patterns, what might be a suitable handling strategy, assuming a large dataset with diverse age groups?
   a) Drop all rows with missing age (potential data loss if many).
   b) Fill with a constant like -1 (might mislead the model).
   c) **Fill with the mean or median age of the entire student population.**
   d) Use One-Hot Encoding on the missing value.

6. In Tutorial Question 02, the director wants to group students without any labeled outcomes. The text notes "the dataset is inconsistent and messy." Before applying any clustering algorithm, what mandatory preprocessing tasks are implied by the messy data description?
   a) Model Deployment and Monitoring.
   b) Feature Selection and Hyperparameter Tuning.
   c) **Handling missing data, converting categorical data to numerical, and scaling numerical features.**
   d) Cross-Validation and Overfitting detection.

7. For the given dataset in Tutorial Question 01, if `Age` is 18 for most students but a few are 40+, which data cleaning problem would this represent if an entry for age was mistakenly "180"?
   a) Incomplete data

b) Inconsistent data
c) Intentional missing data
d) **Noisy data (an error or outlier)**